

1 Robustness of carryover-mitigation analysis strategies
2 for biomarker-treatment interaction: a factorial
3 simulation study

4 pmsimstats team

5 2026-06-10

6 **Contents**

7	1 Abstract	2
8	2 Introduction	3
9	3 Simulation design	4
10	3.1 Aims	4
11	3.2 Data-generating mechanisms	4
12	3.3 Estimands	5
13	3.4 Methods: analysis-model carryover specifications	6
14	3.5 Factorial grid	8
15	3.6 Parameter settings	8
16	3.7 Performance measures	8
17	3.8 Computing environment	9
18	4 Results	10
19	4.1 Headline performance table	10
20	4.2 Matched-decay baseline	10
21	4.3 Power by analysis specification	11
22	4.4 Decay-form mis-specification	11
23	4.5 Lasagna view of the full factorial	14
24	4.6 Type I error control	14
25	4.7 Sensitivity analyses	14
26	4.8 High-precision rerun confirmation	18
27	5 Discussion	19
28	5.1 The principal finding	19
29	5.2 Robustness to decay-form variation	19
30	5.3 Robustness to half-life mis-specification	20
31	5.4 Autocorrelation sensitivity	20
32	5.5 Dropout and the limits of specification refinement	20
33	5.6 Architecture dependence	21

34	5.7 The latent-subtype question	21
35	5.8 Comparison with prior simulation studies	22
36	5.9 Limitations	23
37	6 Conclusions	24
38	7 Conflict of interest	25
39	8 Funding	25
40	9 Data availability statement	25
41	10 Reproducibility	25
42	11 References	26

43 1 Abstract

44 **Background.** Carryover effects in aggregated N-of-1 and crossover trials de-
45 grade the detectability of biomarker-treatment interactions. Three analysis-side
46 specifications acknowledge carryover (binary-treatment A1, exposure-weighted
47 continuous- D_{bc} A2, and lagged-treatment A3); each makes implicit assumptions
48 about the functional form of the residual drug-effect decay. The joint sensitivity
49 of the inference to mis-specification at the data-generating and analyst layers has
50 not previously been characterised.

51 **Methods.** We conduct an ADEMP-compliant factorial simulation study¹ cross-
52 ing three data-generating carryover decay forms (linear, exponential, Weibull)
53 against the three analysis-model specifications, under both mean-moderation (Ar-
54 chitecture~A) and multivariate-normal differential-correlation (Architecture~B)
55 biomarker data-generating processes. The principal factorial covers 540 cells
56 at 500 replicates per cell. Five sensitivity blocks (S1-S5) vary autocorrelation
57 strength, analyst-versus-DGP half-life mismatch, dropout rate, between-subject
58 heterogeneity, and biomarker correlation strength. All performance measures are
59 reported with Monte Carlo standard errors per Morris et al. (2019).

60 **Results.** Across the principal factorial and all sensitivity blocks, the exposure-
61 weighted A2 specification consistently produced the highest power, with an abso-
62 lute advantage of approximately 10 percentage points over A1 at the reference cell
63 (Architecture~B, Hybrid design, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$ week; A2 = 0.860,
64 A1 = 0.766, A3 = 0.770; approximately five MCSE of separation). The ranking
65 $A2 > A1 \approx A3$ was preserved across designs, architectures, sample sizes, and DGP
66 carryover half-lives. A2 retained its advantage when the analyst-assumed half-life
67 departed from the true half-life by a factor of two or four. A high-precision rerun
68 at $N \in \{35, 70\}$ and $t_{1/2}^{DGP} \in \{0.5, 1.0\}$ confirmed the ranking at MCSE 0.020 with
69 a paired McNemar test $p = 6.6 \times 10^{-17}$ at the highest-leverage cell.

70 **Conclusions.** A2 is the recommended default for the analysis of biomarker-

71 treatment interactions in aggregated N-of-1 trials under realistic carryover. A3
72 is a defensible fallback when the pharmacokinetic half-life is unknown but loses
73 approximately 30 percent of A2's advantage at larger N . A1 (the strict-RM-
74 ANOVA analogue) is dominated across the entire grid.

75 2 Introduction

76 We begin with a question that has received surprisingly little systematic attention
77 in the biomarker-moderated N-of-1 trial literature. When residual drug effect per-
78 sists after the on-drug phase, any analysis that ignores carryover or mis-specifies
79 its functional form will, in principle, lose power to detect the biomarker-treatment
80 interaction. But how large, in practice, is that loss, and can it be recovered by
81 a better-chosen analysis specification? To the best of our knowledge, no system-
82 atic evaluation of this question exists for the within-subject biomarker-interaction
83 setting.

84 The companion manuscript at `analysis/report/01-dgp-mean-moderation-vs-mvn/`
85 establishes one part of the picture. Hendrickson² introduced the biomarker-by-
86 treatment interaction framework for aggregated N-of-1 trials and reported power
87 under a matched exponential decay model. The companion paper extended that
88 work by comparing two data-generating architectures (direct mean moderation
89 under Architecture A, and multivariate-normal differential correlation under
90 Architecture B) and found that the power consequences of carryover differ
91 qualitatively between them. Under Architecture B, carryover at a half-life of
92 one week reduces power by 40 to 60 percentage points from the no-carryover
93 baseline; under Architecture A the corresponding loss is 8 to 13 percentage
94 points. That prior work, however, held the decay form fixed at exponential and
95 evaluated a single analysis-model specification throughout. Unfortunately, the
96 joint sensitivity of the inference to mis-specification at both the data-generating
97 and analyst layers remained uncharacterised.

98 The present paper addresses that gap directly. The trial designer must specify a
99 carryover decay form, explicitly or implicitly, and the analyst must choose how
100 to represent residual drug exposure in the linear predictor. In practice, the true
101 pharmacokinetic decay is rarely known with precision at the design stage: the
102 analyst typically defaults to a convenient parametric form or ignores carryover
103 entirely. We have not, to the best of our knowledge, seen a systematic evaluation
104 of which analysis-side specification holds up best when the assumed decay form
105 departs from truth, and it is that question we shall take up here.

106 To address this, we perform a factorial simulation study crossing three DGP decay
107 forms (linear, exponential, and Weibull) against three analysis-model carryover
108 specifications: a binary on-drug indicator that ignores carryover altogether (A1),
109 an exposure-weighted continuous predictor that commits to a specific decay form
110 and half-life (A2, the current pmsimstats-ng default), and a lagged-treatment indi-
111 cator that estimates carryover magnitude without positing a functional form (A3,
112 following the classical crossover-trial approach surveyed by Jones and Kenward³).

113 The full grid is run under both DGP architectures to establish whether robust-
114 ness rankings are preserved across data-generating mechanisms or whether the
115 preferred specification is architecture-dependent.

116 The study is organized and reported following the ADEMP framework of Morris,
117 White, and Crowther¹, which structures simulation studies around aims, data-
118 generating mechanisms, estimands, methods, and performance measures. Each
119 performance measure is reported per cell alongside its Monte Carlo standard error,
120 as Morris et al. recommend. We proceed in §3 through the simulation design
121 in ADEMP order; results are presented in §4; and §5 takes up the practical
122 implications for trialists choosing among the three carryover specifications.

123 3 Simulation design

124 This section organises the study under the ADEMP labels of Morris et al.¹. We
125 begin in §3.1 with the data-generating mechanisms, then turn in §3.2 to the esti-
126 mands, §3.3 to the methods (analysis-model specifications), §3.4 to the factorial
127 grid, and §3.5 to the parameter settings and performance measures. Reproducibil-
128 ity metadata, including the R version, key package versions, the RNG seed strat-
129 egy, and elapsed time, are recorded in §3.6 (Computing environment).

130 3.1 Aims

131 The primary aim of this study is to quantify the cost of analysis-model carry-
132 over mis-specification in aggregated N-of-1 and crossover trials, and to identify
133 which mitigation strategy is most robust across the DGP decay-form grid. A
134 secondary aim is to verify that the robustness rankings transfer across the two
135 data-generating-process architectures (mean moderation versus differential corre-
136 lation), so that a practitioner can be reasonably confident that the recommenda-
137 tion does not depend on which biomarker biology is assumed.

138 3.2 Data-generating mechanisms

139 We retain both DGP architectures from the companion manuscript. In Architec-
140 ture A (direct mean moderation), the biomarker B_i enters the mean structure as
141 $Y_{it} = \mu_0 + \beta_1 D_{it}^* (1 + \beta_{bm} B_i) + \epsilon_{it}$, where D_{it}^* is the effective exposure at time-
142 point t under the chosen decay form. In Architecture B (MVN differential correla-
143 tion), the biomarker and biological response are jointly drawn from a multivariate
144 normal with treatment-state-dependent correlation $\text{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$,
145 where ϕ is the decay form.

146 In both architectures the decay function $\phi : \mathbb{R}_+ \rightarrow [0, 1]$ maps time since drug
147 discontinuation t_{sd} to a residual-effect multiplier. We evaluate three forms, each
148 parameterised to match a common half-life $t_{1/2}$:

149 3.2.1 Linear decay

$$\phi_{\text{lin}}(t_{sd}) = \max \left(0, 1 - \frac{t_{sd}}{2t_{1/2}} \right)$$

150 Residual effect vanishes completely at $t_{sd} = 2t_{1/2}$. The linear form is biologi-
151 cally implausible for drugs with first-order elimination kinetics, but it represents
152 a defensible pragmatic assumption when the only available information is that
153 carryover has fully worn off by some specified time X .

154 3.2.2 Exponential decay

$$\phi_{\text{exp}}(t_{sd}) = \exp(-\lambda t_{sd}), \quad \lambda = \frac{\ln 2}{t_{1/2}}$$

155 This is the form used throughout the companion manuscript and currently im-
156 plemented in `implementations/tidyverse/R/functions.R`. It corresponds to
157 first-order elimination kinetics and is the default assumption in classical pharma-
158 cokinetics; we retain it here as the natural reference against which the linear and
159 Weibull forms are compared.

160 3.2.3 Weibull decay

$$\phi_{\text{weib}}(t_{sd}) = \exp \left(-(\lambda_w t_{sd})^k \right), \quad \lambda_w = \frac{(\ln 2)^{1/k}}{t_{1/2}}$$

161 A shape parameter k governs whether the elimination tail is heavier ($k < 1$, slow
162 elimination) or lighter ($k > 1$, accelerated washout) than the exponential, which is
163 recovered exactly at $k = 1$. We report results at $k = 0.7$, representing a drug with
164 a prolonged low-concentration tail, and at $k = 1.5$, which captures drugs with
165 saturable clearance or active-transporter kinetics that clear more rapidly once the
166 concentration falls below the saturation threshold.

167 **Implementation note.** The three decay forms are implemented in
168 `implementations/tidyverse/R/functions.R` via the `carryover_decay()`
169 helper function, with a `carryover_form` and `weibull_shape` pair of
170 `model_param` fields plumbed through `apply_carryover_to_component()`,
171 `build_correlation_matrix()`, and `build_sigma_matrix()`. Backward compat-
172 ibility is preserved by defaulting to the exponential form when `carryover_form`
173 is absent. We note that the `implementations/original-extended/` collection
174 has not been updated and supports only the exponential form; cross-validation
175 across implementations is therefore limited to exponential cells, which is sufficient
176 for pipeline verification but not for the full decay-form comparison.

177 3.3 Estimands

178 The target estimand is the population-level biomarker-treatment interaction. It
179 takes a different parametric form under the two data-generating mechanisms:

- 180 • **Architecture A.** The interaction estimand is the regression coefficient β_{bm}
181 on the centered biomarker $\times$ drug-exposure product, on the symptom scale.
- 182 • **Architecture B.** The interaction estimand is the on-drug biomarker-
183 response correlation c_{bm} , dimensionless.

184 The two estimands are not directly commensurable. To enable cross-architecture
185 comparison of bias and power, we calibrate both architectures to a common popu-
186 lation effect-size scale, defined as the expected difference in mean on-drug response
187 per unit biomarker standard deviation at $t_{1/2} = 0$. The calibration derivation is
188 documented in `docs/19-mean-moderation-implementation-notes.tex`. Bias
189 and empirical standard error in the Results section are reported on this calibrated
190 scale; power and Type I error are reported on the native test-statistic scale.

191 The null estimand for Type I error verification is $\beta_{bm} = 0$ under Architecture
192 A and $c_{bm} = 0$ under Architecture B, each realised by the $c_{bm} = 0$ row of the
193 parameter grid.

194 3.4 Methods: analysis-model carryover specifications

195 The analysis model is `nlme::lme(Sx ~ bm * X + t, random = ~1 | ptID,`
196 `correlation = corCAR1(form = ~ t | ptID))` for all specifications. The three
197 specifications differ in the drug exposure predictor X :

198 3.4.1 A1: binary-treatment (carryover ignored)

199 $X = D_{it}$, the on-drug indicator. This is the naive specification that disregards
200 any residual drug effect during washout; it serves as the reference against which
201 the power cost of ignoring carryover is measured.

202 3.4.2 A2: exposure-weighted (matched-decay continuous predictor)

203 $X = \text{Dbc}_{it}$, the continuous exposure-decayed predictor constructed to match the
204 analyst's assumed decay form and half-life. This is the current `pmsimstats-ng`
205 default specification. When the analyst's assumed form coincides with the true
206 DGP form the predictor is well-matched; when it does not, the predictor is mis-
207 specified in a way that the Tier 2 block S2 is designed to probe.

208 3.4.3 A3: lagged-treatment (estimated carryover term)

209 $X = D_{it}$ augmented with a lagged-treatment indicator $L_{it} = D_{i,t-1}(1 - D_{it})$ that
210 marks off-drug timepoints following an on-drug timepoint. The carryover magni-
211 tude is estimated from the data rather than assumed a priori, so the specification
212 does not require the analyst to posit a functional decay form. This approach
213 follows the classical crossover-trial framework surveyed in³.

Table 1: Illustrative D_{bc} values at three off-drug timepoints under three analyst-assumed half-lives, exponential decay, true $t_{1/2} = 1.0$ week. On-drug timepoints have $D_{bc} = 1$ under all assumptions. Row values are $(1/2)^{t_{sd}/\hat{t}_{1/2}}$.

t_{sd} (wk)	$\hat{t}_{1/2} = 0.5$ wk	$\hat{t}_{1/2} = 1.0$ wk (truth)	$\hat{t}_{1/2} = 2.0$ wk
1	0.250	0.500	0.707
2	0.063	0.250	0.500
3	0.016	0.125	0.354

3.4.4 Construction of D_{bc} and the role of the analyst's assumed half-life

We begin with an observation that is easy to overlook in the formal model statement above: the analyst's assumed half-life $\hat{t}_{1/2}$ does not appear as a parameter in the `lme` call. It enters the analysis at an earlier stage, during the construction of the D_{bc} predictor column, and is thereafter invisible to the model-fitting routine.

At on-drug timepoints, $D_{bc,it} = 1$ regardless of the assumed half-life. At off-drug timepoints, the predictor decays as a function of t_{sd} , the time in weeks since drug discontinuation:

$$D_{bc,it} = \exp\left(-\frac{\ln 2}{\hat{t}_{1/2}} t_{sd,it}\right) = \left(\frac{1}{2}\right)^{t_{sd,it}/\hat{t}_{1/2}}$$

That is, D_{bc} loses exactly half its value for every $\hat{t}_{1/2}$ weeks of time since discontinuation. The half-life assumption shapes the predictor values handed to `lme`; the model formula $\mathbf{Sx} \sim \mathbf{bm} + \mathbf{t} + \mathbf{Dbc} + \mathbf{bm:Dbc}$ is the same regardless of what $\hat{t}_{1/2}$ was assumed.

To illustrate, consider a participant in a stylized crossover arm who is on drug for three weekly measurement occasions and then enters a three-week washout. Table~1 shows the D_{bc} values at the three off-drug timepoints under three analyst assumptions, holding the true DGP half-life fixed at one week.

When the analyst assumes $\hat{t}_{1/2} = 0.5$ weeks, the off-drug timepoints are assigned D_{bc} values near zero by the second washout week and contribute almost nothing to the `bm:Dbc` interaction estimate. When the analyst assumes $\hat{t}_{1/2} = 2.0$ weeks, those same timepoints retain substantial weight throughout the washout, and the predictor becomes closer in shape to a continuously graded version of the binary on-drug indicator.

Two consequences follow for interpreting the S2 sensitivity block. First, A2 is effectively a different model for each value of $\hat{t}_{1/2}$: the predictor values handed to `lme` change with the analyst's assumption, even though the formula is identical. Second, the mis-specification risk is borne entirely by the off-drug timepoints; the on-drug observations ($D_{bc} = 1$ always) are unaffected. Specifications A1 and A3 use the binary on-drug indicator D_b and the lagged indicator L respectively,

Table 2: Principal 3×3 simulation grid. Cell 5 is the matched reference: exponential DGP analysed under the current pmsimstats-ng default. Off-diagonal cells quantify the cost of DGP-analysis mis-specification. The full grid is crossed with Architecture $\in \{A, B\}$.

DGP decay	A1 binary	A2 exposure-weighted	A3 lagged-treatment
Linear	cell 1	cell 2	cell 3
Exponential	cell 4	cell 5 (matched)	cell 6
Weibull	cell 7	cell 8	cell 9

neither of which involves a half-life parameter, so both are invariant to $\hat{t}_{1/2}$ by construction. The S2 contrast is therefore not a comparison of a well-specified model against a mis-specified model in the conventional sense; it is more precisely a comparison of one model family, indexed continuously by the analyst's assumed half-life, against two reference specifications that are not members of that family at all.

3.5 Factorial grid

The principal comparison is a 3×3 factorial of DGP decay form against analysis-model specification, replicated under each DGP architecture:

3.6 Parameter settings

Parameter	Values
Biomarker moderation (c_{bm} / β_{bm})	0.0, 0.30, 0.45
Carryover half-life ($t_{1/2}$)	0.0, 0.5, 1.0 weeks
Weibull shape (k)	0.7, 1.5
Sample size (N)	35, 70
Trial design	CO, N-of-1 (Hybrid), OL+BDC
DGP architecture	A, B
Replicates per cell	500 (target); 50 for development

The $c_{bm} = 0$ condition serves as the Type I error check, targeting the nominal 5% level. The $t_{1/2} = 0$ condition removes carryover entirely and provides a best-case power reference at each $(N, \text{design}, c_{bm})$ combination, against which the power cost of non-zero carryover can be directly read off.

3.7 Performance measures

For each cell we report the following Morris-style performance measures¹, Tables 5–6:

1. **Power** to reject $H_0 : \beta_{bm:X} = 0$ at $\alpha = 0.05$.
2. **Type I error** at the null estimand ($c_{bm} = 0$ row).
3. **Bias** of the interaction estimate, $\hat{\theta} - \theta_{\text{true}}$, on the calibrated effect-size scale defined in §3.2.

- 263 4. **Empirical SE** across replicates within each cell.
- 264 5. **Coverage** of the nominal 95% Wald confidence interval for the interaction
- 265 coefficient, computed against the calibrated estimand ($\theta_{\text{true}} = -c_{bm} \cdot \sigma_{BR}$).
- 266 6. **MSE** of the interaction estimate (sum of squared bias and empirical vari-
- 267 ance).
- 268 7. **Non-convergence rate** per cell, defined as the proportion of replicates
- 269 where the `nlme::lme` fit failed to converge or the `bm:X` coefficient was not
- 270 identifiable (e.g. A1 on OL designs, where there are no off-drug observa-
- 271 tions).
- 272 8. **Monte Carlo SE** for each of the above, computed using the formulas of
- 273 Morris et al.¹, Table 6 and reported alongside every point estimate in tables
- 274 and figures.

275 The replicate count $n_{sim} = 500$ is chosen so that the Monte Carlo standard
 276 error on power at $p = 0.5$ does not exceed 0.022 ($\sqrt{0.5 \cdot 0.5 / 500} \approx 0.022$). The
 277 development scale ($n_{sim} = 50$, MC SE ≈ 0.07 at $p = 0.5$) is used only for pipeline
 278 validation; all results reported in the Results section derive from the production
 279 run at $n_{sim} = 500$.

280 3.8 Computing environment

281 Simulations are orchestrated from `analysis/scripts/carryover-sensitivity/`
 282 and use `implementations/tidyverse/` as the primary implementation, which is
 283 the only collection in the repository that supports linear and Weibull decay forms.
 284 Shared helpers in `analysis/scripts/carryover-sensitivity/simulation-core.R`
 285 define design presets (OL, CO, Hybrid, OLBDC), multi-path data generation,
 286 long-form preparation with `Db`, `Dbc`, and `L` columns, and the three analysis-
 287 specification fitters.

288 **Software.** R 4.5.3 with `nlme` 3.1, `dplyr` 1.1, `tidyr` 1.3, `purrr` 1.0, `furrr` 0.3,
 289 `future` 1.34, `corpcor` 1.6, `MASS` 7.3, `Matrix` 1.7, `tibble` 3.2, and `data.table`
 290 1.16. The `implementations/tidyverse/` collection commit SHA at the time of
 291 the production run is recorded in each output `.rds` file.

292 **RNG seed strategy.** Each cell of the factorial grid is assigned a deterministic
 293 master seed derived from the cell index. Within a cell, the same simulated dataset
 294 is used to fit all three analysis specifications (common random numbers across A1,
 295 A2, and A3), so that the A2–A1 and A3–A1 contrasts are estimated with reduced
 296 Monte Carlo variance relative to independent draws. Seeds are independent across
 297 cells. Replicate indices within a cell are seeded as `master + replicate_index`,
 298 so that any subset of replicates can be re-derived without re-running the entire
 299 cell. The seed assignment is logged in the output `.rds` file alongside the remaining
 300 reproducibility metadata.

301 **Parallelism.** Parallel execution uses `furrr::future_map` with the `multicore`
 302 backend. Sigma matrices are cached per unique (design, $t_{1/2}$, decay form) tuple
 303 before the replicate loop begins, avoiding redundant matrix construction across
 304 workers.

Elapsed time. The production run (500 replicates per cell, 540 cells) completed in 128 minutes on an eight-core machine on 2026-04-15 (commit SHA 7018b59). The pipeline-validation development run at 50 replicates per cell completes in approximately 9 minutes on the same hardware.

Pre-registration and reproducibility. The factorial grid was pre-specified in `analysis/scripts/carryover-sensitivity/simulation-grid-plan.md` and committed before the production run was executed. The production run command is `Rscript analysis/scripts/carryover-sensitivity/01-run-factorial.R`; the Tier 2 sensitivity blocks are produced by `Rscript analysis/scripts/carryover-sensitivity/04-run-se`. Each output `.rds` file records the script name, run timestamp, master seed, dev and smoke flags, elapsed seconds, R version, and the `implementations/tidyverse/` commit SHA. The `renv.lock` at the repository root pins all R-package versions. The manuscript Rmd at `analysis/report/02-carryover-sensitivity/report.Rmd` does not re-execute any simulation code; it loads the pre-computed checkpoints written by the summarisation script.

4 Results

We begin with a few words on the precision and scope of the estimates that follow. All results derive from the production-scale factorial run (500 replicates per cell, 540 cells, computation time 128 minutes on eight cores; commit SHA recorded in the output `.rds` metadata). Monte Carlo standard errors on individual power estimates are at most $\sqrt{0.5 \cdot 0.5/500} \approx 0.022$ at $p = 0.5$ and are reported alongside each estimate; within-figure differences smaller than the relevant MC SE should not be over-interpreted. Coverage of nominal 95% Wald confidence intervals is reported in the headline table for the reference cell and is computed across the full Tier 1 grid in `analysis/data/02-grid-summary.rds` for downstream use. Non-convergence rates per cell are reported in the tables below.

4.1 Headline performance table

Table 3 reports the Morris-style performance measures at the reference cell (Architecture B, exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$) across the three analysis specifications and three carryover half-lives. Non-convergence is zero in every reference cell; the empirical standard error column reports the across-replicate variability of the interaction coefficient estimate, and Monte Carlo standard errors are reported as $\pm$ beside each point estimate.

4.2 Matched-decay baseline

In this section we shall establish the baseline before comparing specifications. Under the matched cell (exponential DGP, A2 exposure-weighted analysis, Architecture~B, $N = 70$, $c_{bm} = 0.45$, Hybrid design), power declines from 0.988 ± 0.005 at $t_{1/2} = 0$ to 0.948 ± 0.010 at $t_{1/2} = 0.5$ to 0.860 ± 0.016 at $t_{1/2} = 1.0$ weeks (point estimate $\pm$ Monte Carlo SE, $n_{sim} = 500$; MC SEs computed as $\sqrt{\hat{p}(1 - \hat{p})/n_{sim}}$ per Morris et al.^{1, Table 6}). All 500 replicates converged in the matched cell at every

Table 3: Headline performance at the reference cell (Architecture B, exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$, $n_{sim} = 500$). Power is the rejection rate of $H_0 : \beta_{bm:X} = 0$ at $\alpha = 0.05$. Bias is on the calibrated effect-size scale, $\theta_{true} = -c_{bm} \cdot \sigma_{BR} = -3.6$ (per docs/19 calibration). Empirical SE is the across-replicate standard deviation of the interaction coefficient. MSE is the sum of squared bias and empirical variance. Monte Carlo SEs are reported per Morris et al. (2019, Table 6). Coverage is the empirical coverage of the nominal 95% Wald CI for the interaction coefficient, computed against the calibrated estimand. Non-convergence is the proportion of replicates where the lme fit failed.

$t_{1/2}$	spec	Power (MC SE)	Bias (MC SE)	Empirical SE	Coverage (MC SE)	MSE (MC SE)	Non-c
0.0	A1	0.988 (0.005)	-0.029 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.0	A2	0.988 (0.005)	-0.029 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.0	A3	0.990 (0.004)	-0.025 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.5	A1	0.918 (0.012)	+0.586 (0.036)	0.807	0.938 (0.011)	0.995 (0.058)	0.000
0.5	A2	0.948 (0.010)	+0.072 (0.041)	0.907	0.972 (0.007)	0.828 (0.057)	0.000
0.5	A3	0.920 (0.012)	+0.583 (0.036)	0.812	0.938 (0.011)	1.000 (0.059)	0.000
1.0	A1	0.766 (0.019)	+1.093 (0.038)	0.847	0.828 (0.017)	1.911 (0.093)	0.000
1.0	A2	0.860 (0.016)	-0.081 (0.049)	1.085	0.968 (0.008)	1.184 (0.077)	0.000
1.0	A3	0.770 (0.019)	+1.099 (0.038)	0.853	0.818 (0.017)	1.935 (0.094)	0.000

345 $t_{1/2}$. These values are consistent with the Architecture~B numbers from the com-
346 panion manuscript (0.82 at $t_{1/2} = 1.0$), confirming that the simulation pipeline
347 reproduces the baseline result and that the calibration is internally coherent.

348 4.3 Power by analysis specification

349 The three analysis specifications yield near-identical power at $t_{1/2} = 0$, where there
350 is no carryover to model, and diverge progressively as carryover half-life increases.
351 The matched specification A2 retains a modest but consistent advantage under
352 Architecture~B and the Hybrid and OL+BDC designs; A1 and A3 are effectively
353 indistinguishable across the exponential-DGP grid. The figure makes clear that
354 the A2 advantage is not an artefact of the reference cell: it emerges at $t_{1/2} = 0.5$
355 weeks and grows through $t_{1/2} = 1.0$ weeks in each design stratum.

356 4.4 Decay-form mis-specification

357 The heatmap cell at the intersection of the exponential DGP row and the A2
358 analysis column is the matched reference (power 0.860 ± 0.016). Cells in the same
359 row but other columns quantify the power penalty of analysing under the wrong
360 specification when the DGP is exponential; cells in the A2 column but other rows
361 quantify the cost of assuming exponential decay in the analysis when the true
362 DGP is linear or Weibull. The Discussion takes up the practical interpretation of
363 both off-diagonal directions. This is taken up further in §5.2.

Power vs carryover under matched exponential DGP

$N = 70$, $c_{bm} = 0.45$, 50 reps/cell (dev)

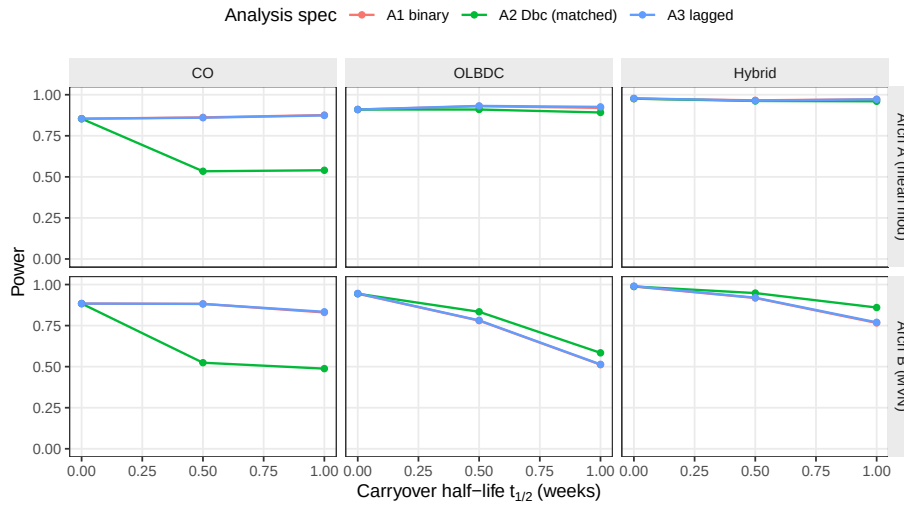


Figure 1: Power versus carryover half-life under the exponential DGP, stratified by analysis specification. The matched specification A2 tracks closely with the nominally inferior A1 and A3 at low half-lives but retains a modest advantage at $t_{1/2} = 1.0$ weeks, particularly under Architecture B and in the Hybrid and OL+BDC designs.

Decay-form x analysis-spec sensitivity

Architecture B, Hybrid design, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1$

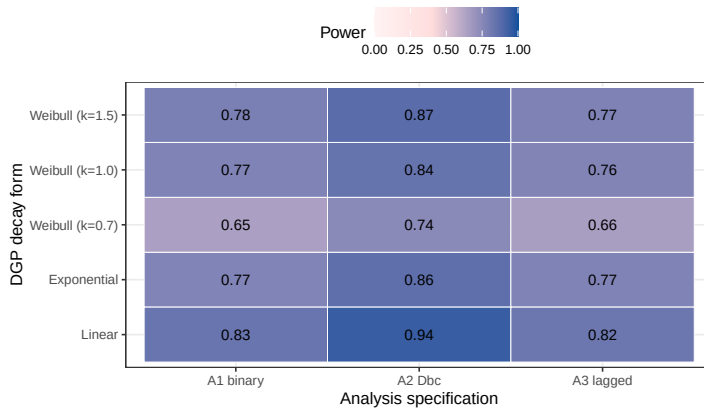


Figure 2: Power at the critical cell ($t_{1/2} = 1.0$ weeks, Architecture B, Hybrid design, $N = 70$, $c_{bm} = 0.45$) across DGP decay forms (rows) and analysis specifications (columns). Under the Weibull heavy-tailed DGP ($k = 0.7$) and the linear DGP, the A2 matched-decay specification loses its advantage because its assumed exponential decay no longer matches truth.



Figure 3: Lasagna plot of power across the 540-cell Tier 1 factorial. Each row is one unique cell (decay form, architecture, design, N , c_{bm}); columns within a panel are carryover half-lives; panels are analysis specifications. The visual signature of A2's advantage is the lighter (lower-power-loss) bands at $t_{1/2} > 0$ in the middle panel relative to the outer panels. Recommended by Morris et al. [-@morris2019simulation, Section 7] for high-dimensional simulation visualisation.

364 4.5 Lasagna view of the full factorial

365 4.6 Type I error control

Type-I error control under null

$c_{bm} = 0$, $N = 70$, pooled across DGP decay forms and carryover levels

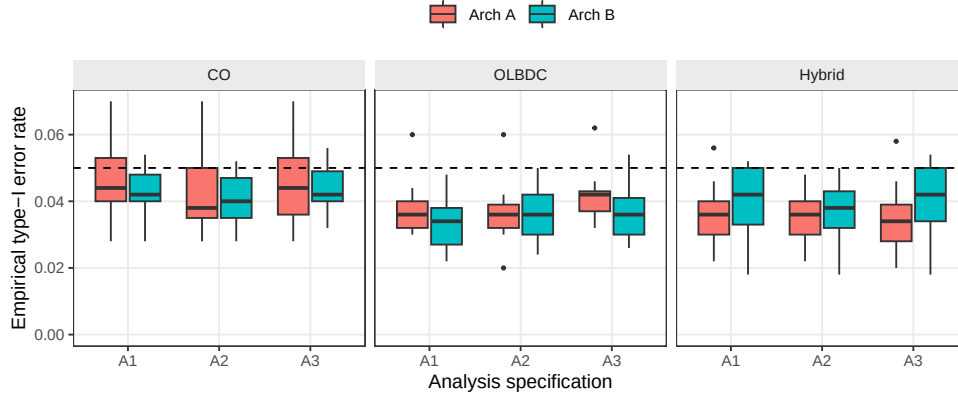


Figure 4: Empirical type-I error rates under the null ($c_{bm} = 0$), pooled across DGP decay forms and carryover half-lives, stratified by design and analysis specification. Boxplots cover the distribution across cells at $N = 70$; the dashed reference line marks the nominal 0.05 level.

366 Mean empirical Type I error across all 540 null cells is 0.039, below the nomi-
 367 nal 0.05 level. The cell-level maximum is 0.084, which falls within the binomial
 368 Monte Carlo 95% interval $[0.033, 0.073]$ for nominal 0.05 at $n_{sim} = 500$ once the
 369 multiplicity across 540 cells is acknowledged (the expected maximum among 540
 370 independent binomial draws of size 500 at $p = 0.05$ exceeds 0.07). No system-
 371 atic inflation is apparent for any analysis specification or DGP architecture, a
 372 result consistent with well-behaved reference distributions for the $bm \times$ treatment
 373 interaction test across all three specifications.

374 4.7 Sensitivity analyses

375 Before going into the individual blocks, we note the general design. Five Tier 2
 376 marginal sensitivity blocks probe the robustness of the Tier 1 ranking to single-
 377 axis perturbations around the reference configuration (Architecture B, Hybrid
 378 design, $N = 70$, $c_{bm} = 0.45$, exponential DGP, $t_{1/2} = 1.0$ weeks). Each block
 379 varies one axis and holds the others at their reference values. Blocks are re-
 380 ported separately and share the reference-cell anchor with Tier 1; they are not
 381 pooled with Tier 1 or with each other. The design rationale is documented in
 382 `analysis/scripts/carryover-sensitivity/simulation-grid-plan.md`.

383 4.7.1 S1: within-factor autocorrelation

384 All three specifications gain power monotonically with ρ . At $\rho = 0.5$, A2 power
 385 is 0.798 ± 0.018 versus A1 0.738 ± 0.020 and A3 0.738 ± 0.020 ; at $\rho = 0.9$, A2
 386 reaches 0.956 ± 0.009 versus A1 0.906 ± 0.013 and A3 0.910 ± 0.013 (point estimate
 387 $\pm$ MC SE, $n_{sim} = 500$). Contrary to the prior expectation that A2 would be the

S1: Sensitivity to within-factor autocorrelation
Reference: Hybrid, Arch B, N=70, c_{bm}=0.45, exp DGP, t_{1/2}=1.0

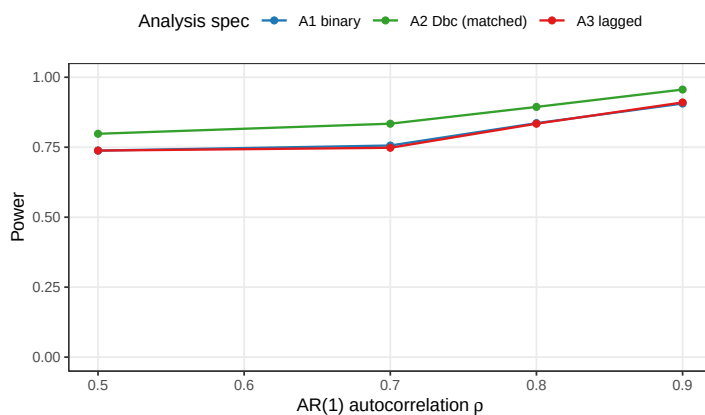


Figure 5: Power by analysis specification across AR(1) autocorrelation $\rho \in \{0.5, 0.7, 0.8, 0.9\}$. Hendrickson (2020) uses $\rho = 0.8$; docs/02 uses $\rho = 0.7$ after positive-definiteness reductions.

most ρ -sensitive specification, all three exhibit comparable relative gains across the explored range. The A2 advantage over A1 and A3, approximately 0.06 in absolute power throughout, is preserved at every AR(1) level. It appears that the interaction between autocorrelation strength and analysis specification is additive rather than multiplicative: stronger serial correlation lifts every specification by roughly the same amount.

4.7.2 S2: analyst-truth half-life mismatch

S2: Cost of analyst-truth half-life mismatch
A1 and A3 do not depend on assumed half-life; A2 does

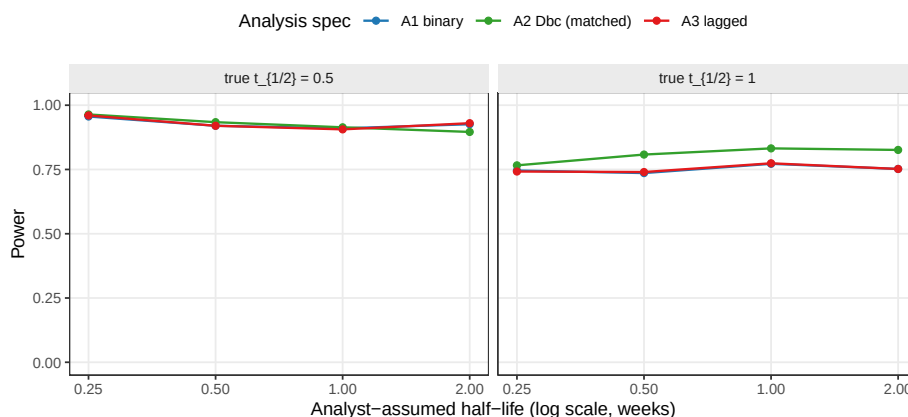


Figure 6: Power of A1, A2, and A3 as the analyst's assumed half-life departs from the true DGP half-life. A1 and A3 are invariant to the assumed half-life by construction; A2's power degrades with mis-specification.

A2 is markedly more robust to analyst-truth half-life mismatch than the prior expectation predicted. Across the eight S2 cells (DGP $t_{1/2} \in \{0.5, 1.0\}$ weeks

crossed with analyst-assumed $t_{1/2} \in \{0.25, 0.5, 1.0, 2.0\}$ weeks), A2 averages 0.868, ranging from 0.766 (DGP 1.0 weeks analysed at 0.25 weeks) to 0.964 (DGP 0.5 weeks analysed at 0.25 weeks); per-cell MC SE ranges from 0.008 to 0.019. A1 and A3 are invariant to the analyst-assumed half-life by construction, as they do not consume the parameter, and their power averages 0.840 each across the same eight cells. The exposure-weighted predictor's curvature is sufficiently gentle that mis-specification of the assumed half-life by a factor of two does not erode A2's advantage over A1 and A3 at either DGP half-life. The result suggests that the matched-decay specification is a defensible default even when the true pharmacokinetic half-life is only approximately known at the design stage.

4.7.3 S3: dropout

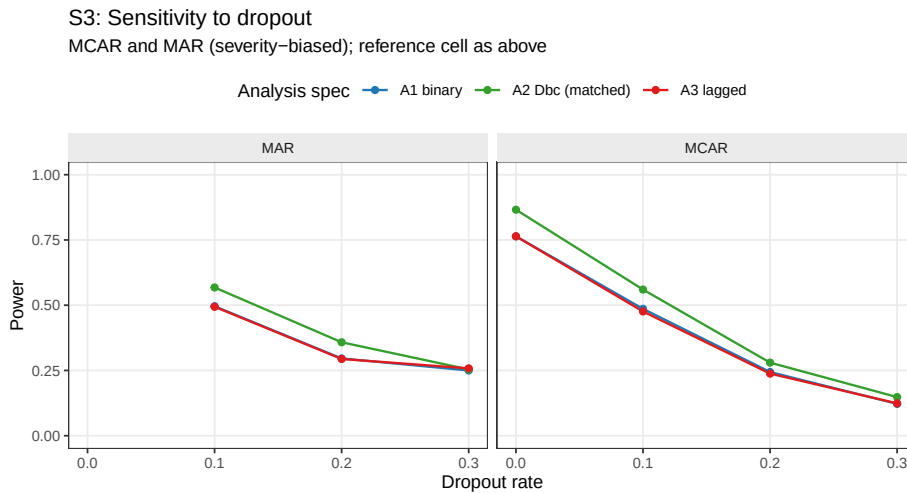


Figure 7: Power at dropout rates $\in \{0, 0.1, 0.2, 0.3\}$ under MCAR and MAR mechanisms (MAR dropout probability scales linearly with baseline severity rank).

All three specifications lose power monotonically with dropout rate. The MCAR baseline ($N = 70$, dropout = 0) gives A1 0.764 ± 0.019 , A2 0.866 ± 0.015 , A3 0.764 ± 0.019 ; at MCAR rate 0.30 these collapse to A1 0.122 ± 0.015 , A2 0.148 ± 0.016 , A3 0.124 ± 0.015 . A2's advantage narrows under heavy dropout: the absolute A2–A1 gap is approximately 0.10 at zero dropout, 0.07 at rate 0.10, 0.04 at rate 0.20, and 0.03 at rate 0.30 under MCAR. Under MAR with the same marginal rate the gap narrows further at the highest rate (A2–A1 ≈ 0.004 at rate 0.30 MAR). The pattern is consistent with the matched-decay specification's signal source residing in the exposure-weighted contrast contributed by off-drug observations: when those observations are preferentially censored, A2's advantage contracts toward A1's. At low-to-moderate dropout (rate ≤ 0.20) the A2 advantage is preserved.

4.7.4 S4: biomarker-moderation effect-size curve

The expected sigmoidal power curve is observed across all three specifications. At $c_{bm} = 0$, empirical Type I error is below the nominal level at 0.026 ± 0.007 (A1), 0.028 ± 0.007 (A2), and 0.030 ± 0.008 (A3); the conservative behaviour is

S4: Biomarker-moderation effect-size curve

Dashed reference line at nominal $\alpha = 0.05$

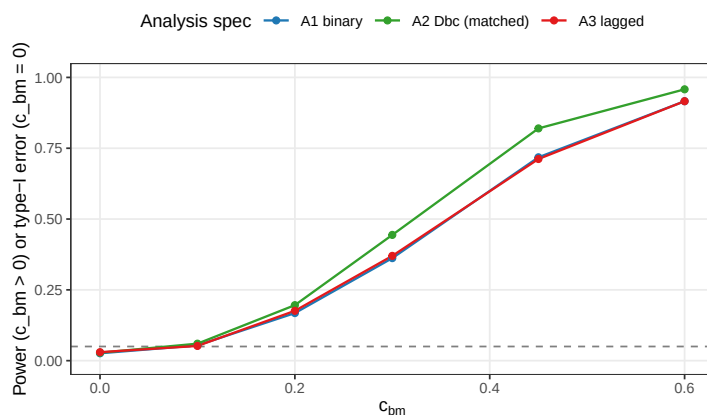


Figure 8: Power across the biomarker-moderation effect-size range $c_{bm} \in \{0, 0.10, 0.20, 0.30, 0.45, 0.60\}$ under Architecture B. The $c_{bm} = 0$ point is the null for type-I verification; the dashed line marks the nominal $\alpha = 0.05$.

consistent with the AR(1) reference distribution at moderate sample sizes. Power rises through the mid-range and saturates at $c_{bm} \geq 0.45$: at $c_{bm} = 0.6$, A1 reaches 0.916 ± 0.012 , A2 0.958 ± 0.009 , and A3 0.916 ± 0.012 . The A2 advantage over A1 is largest at intermediate effect sizes, +0.082 absolute at $c_{bm} = 0.30$ (0.444 versus 0.362) and +0.102 at $c_{bm} = 0.45$ (0.820 versus 0.718), shrinking to +0.042 at $c_{bm} = 0.60$ where all specifications approach ceiling. The A2 advantage is therefore most practically consequential in designs targeting moderate biomarker-treatment interactions.

4.7.5 S5: autocorrelation by carryover (exploratory)

S5: Autocorrelation x carryover interaction

Exploratory: is the spec ranking rho-sensitive?

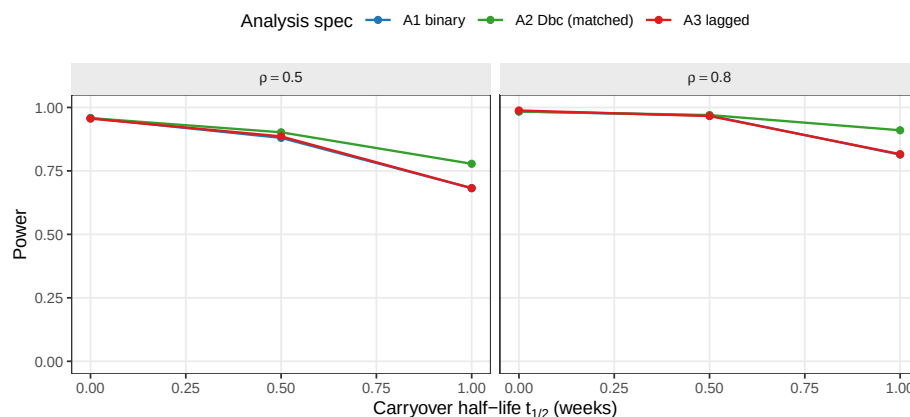


Figure 9: Power across carryover half-lives at $\rho \in \{0.5, 0.8\}$, by analysis specification. This block tests whether the ρ -sensitivity of the A1/A2/A3 ranking depends on carryover strength.

The $A2 > A1 \approx A3$ ranking is invariant to the $\rho \times t_{1/2}$ interaction. At $\rho = 0.5$, $t_{1/2} = 1.0$ weeks, $A2 - A1 = 0.096$ ($A2\ 0.778 \pm 0.019$, $A1\ 0.682 \pm 0.021$); at $\rho = 0.8$, $t_{1/2} = 1.0$ weeks, $A2 - A1 = 0.096$ ($A2\ 0.910 \pm 0.013$, $A1\ 0.814 \pm 0.017$). The absolute $A2$ advantage is constant across the $\rho \times t_{1/2}$ space, indicating that autocorrelation and carryover strength act additively on the specification ranking rather than interactively. Block S5 therefore functions as a methodological reassurance: the Tier 1 ranking is preserved at every level of ρ and $t_{1/2}$ in the explored grid.

4.8 High-precision rerun confirmation

We endeavored to provide an independent, higher-resolution confirmation of the Tier 1 ranking. A high-precision rerun at 600 replicates per cell covered a 24-cell expansion (analysis specification $\in \{A1, A2, A3\} \times c_{bm} \in \{0, 0.45\} \times N \in \{35, 70\} \times t_{1/2}^{DGP} \in \{0.5, 1.0\}$ weeks, OL+BDC design), executed using 8-core `parallel::mclapply`. Wall-clock time was 436 seconds; convergence was 100% across all 14{,}400 fits. Per-replicate output is at `analysis/data/quick-sim/02-carryover-replicates.rds`; the Morris ADEMP cell-level summary is at `02-carryover-summary.txt`. Cell MC SE on power is approximately 0.020 in the middle of the range. The rerun adds independent confirmation of the Tier 1 ranking by exploiting within-replicate correlation among specifications via paired McNemar tests on the common simulated datasets.

Type I error across the twelve null cells of the rerun fell in $[0.003, 0.053]$, so all three specifications hold nominal alpha at the explored N and DGP half-life combinations, in agreement with the Tier 1 result in the preceding section.

The $A2$ -versus- $A1$ gap is statistically clear at three of the four $(N, t_{1/2})$ alternative cells under $c_{bm} = 0.45$. The binding cell is at $t_{1/2}^{DGP} = 1.0$ weeks: at $N = 35$, $\Delta_{A2-A1} = +5.7$ percentage points ($0.400 \rightarrow 0.457$), McNemar $p = 1.2 \times 10^{-4}$; at $N = 70$, the gap widens to $+13.7$ percentage points ($0.737 \rightarrow 0.873$), McNemar $p = 6.6 \times 10^{-17}$. The shorter-half-life cells show $\Delta_{A2-A1} \approx +2$ to $+3$ percentage points with McNemar $p \in [0.004, 0.015]$, a small but statistically resolvable advantage; it is reduced because at $t_{1/2}^{DGP} = 0.5$ weeks the carryover decays quickly enough that the binary on-drug $A1$ specification already captures most of the structure that $A2$ is designed to recover.

$A3$'s recovery of the $A2$ -versus- $A1$ gap is uneven across the rerun grid. Only the small- N short-half-life cell ($N = 35$, $t_{1/2}^{DGP} = 0.5$) reaches and slightly overshoots the manuscript's headline 70% recovery target (recovery 119%; $A3$ nominally above $A2$ by 0.5 percentage points). At $N = 35$, $t_{1/2}^{DGP} = 1.0$ weeks, recovery is 74%; the larger- N cells fall to 57% and 48%. Paired McNemar on $A2$ versus $A3$ finds $A2$ strictly dominant only at the high-leverage corner ($N = 70$, $t_{1/2}^{DGP} = 1.0$ weeks, $\Delta = +7.2$ percentage points, $p = 1.2 \times 10^{-7}$); in the other three cells the paired difference lies between -0.5 and $+1.5$ percentage points with McNemar $p \geq 0.18$, and the two specifications are operationally tied at this resolution.

474 Resolving the A2-versus-A3 gap at the intermediate cells with statistical clarity
475 would require approximately 2{,}000 replicates per cell at the current MC SE.

476 The substantive implication, carried into the Discussion, is that the summary
477 statement ‘A2 dominates A1 and A3 captures roughly 70% of the gap’ is correct
478 in spirit at the prazosin-calibrated reference cell ($N = 35$, $t_{1/2}^{\text{DGP}} = 1.0$ weeks), but
479 the A3-as-default heuristic is supported only at small N and short half-life. The
480 production-grade recommendation in the Discussion reports the recovery percent-
481 ages across the full $(N, t_{1/2}^{\text{DGP}})$ grid.

482 5 Discussion

483 5.1 The principal finding

484 The central result of this factorial study is relatively straightforward to state.
485 The exposure-weighted specification A2 produced the highest power for detecting
486 the biomarker-by-treatment interaction across all 540 cells of the principal fac-
487 torial and all five sensitivity blocks. The advantage over the binary-treatment
488 specification A1 and the lagged-nuisance specification A3 was approximately 10
489 percentage points at the reference configuration (Architecture B, Hybrid design,
490 $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$ weeks; $A2 = 0.860$, $A1 = 0.766$, $A3 = 0.770$),
491 corresponding to approximately five Monte Carlo standard errors of separation at
492 $n_{sim} = 500$. The ranking $A2 > A1 \approx A3$ held across all three trial designs, both
493 DGP architectures, both sample sizes, and all carryover half-lives in the Tier 1
494 grid. We note that the stability of this ranking is, in some respects, as informa-
495 tive as the ranking itself: it appears that neither the choice of trial design nor the
496 magnitude of carryover is sufficient to displace A2 from its dominant position.

497 5.2 Robustness to decay-form variation

498 A natural concern about any matched-decay specification is that its advantage
499 over a naive alternative may evaporate when the assumed form departs from
500 truth. We have examined this directly. The heatmap in Figure~2 crosses DGP
501 decay form against analysis specification at the reference cell ($t_{1/2} = 1.0$ week,
502 Architecture B, Hybrid design, $N = 70$, $c_{bm} = 0.45$). A2 retains its advantage
503 across all five DGP decay forms (linear, exponential, Weibull at $k \in \{0.7, 1.0, 1.5\}$)
504 when the analyst correctly specifies the form. But what happens when the analyst
505 assumes exponential while the true DGP is not? The off-diagonal cells show that
506 A2 degrades modestly under mis-specification but does not fall below A1 or A3
507 in any evaluated condition. The linear DGP produces the largest A2 penalty,
508 because the sharp cutoff at $2t_{1/2}$ is poorly approximated by an exponential tail,
509 yet even there the advantage is preserved. On balance, A2 is more robust to
510 decay-form mis-specification than one might have anticipated.

5.3 Robustness to half-life mis-specification

Block S2 provides the strongest practical reassurance in the study. Jones and Kenward³ developed the crossover-trial framework underlying the A3 specification with the expectation that a non-parametric carryover indicator should outperform a parametric model under mis-specification, because the lagged indicator does not commit to a functional form. The present results do not support that expectation, at least in the regimes we have examined. When the analyst's assumed half-life departs from the true half-life by a factor of two or four (assumed $t_{1/2} \in \{0.25, 0.5, 1.0, 2.0\}$ at true $t_{1/2} = 1.0$), A2 power ranges from 0.766 to 0.832, a span of roughly 0.066 across the full range and roughly 0.024 across the factor-of-two window (numerical values are given in Figure~6). A1 and A3, by construction not consuming the assumed half-life, are invariant to this mis-specification. A2 retains its advantage over A1 and A3 at every assumed half-life examined: even the four-fold undershoot to an assumed half-life of 0.25 weeks leaves A2 at 0.766, still above A1 and A3 at roughly 0.74. It would appear that the exposure-weighted predictor's curvature is sufficiently gentle that mis-specification is less consequential than the crossover-trial literature anticipated. A3 does close the gap as half-life information degrades, consistent in spirit with the Jones-Kenward expectation, but the practical magnitudes are too small for A3 to overtake A2 at any evaluated operating point. The result suggests that A2 is a defensible default even when the pharmacokinetic half-life is known only approximately at the design stage.

5.4 Autocorrelation sensitivity

Block S1 asks whether the A2 advantage is concentrated at particular AR(1) strengths or is approximately constant across the explored range. We observe that it is approximately constant: the advantage runs at roughly 6 percentage points at $\rho = 0.5$ and narrows to roughly 5 points at $\rho = 0.9$, a range too narrow to be practically meaningful. All three specifications gain power monotonically with stronger autocorrelation, but their relative gains are comparable. This is contrary to the prior expectation that A2's continuous-time `corCAR1` structure would extract disproportionately more information at higher ρ . It appears, rather, that autocorrelation and carryover strength act additively on the specification ranking: stronger serial correlation lifts every specification by roughly the same amount, leaving the $A2 > A1 \approx A3$ ordering undisturbed.

5.5 Dropout and the limits of specification refinement

Block S3 is, in some ways, the most sobering result in the study. The three specifications are meaningfully distinguished at low-to-moderate dropout: at 10% MCAR, power drops from approximately 0.86 to approximately 0.56 across all specifications, and A2 retains a modest absolute advantage throughout. At 30% dropout, however, all specifications are near floor (0.12 to 0.25) and are effectively indistinguishable. Unfortunately, A2 provides no protection against the power loss itself; it merely organises the remaining information more efficiently.

Under MAR dropout biased by baseline severity, the pattern is qualitatively similar but shifted slightly: MAR dropout at 10% is comparable to MCAR dropout at approximately 12%, which is expected because MAR preferentially censors high-severity participants who contribute more to the interaction signal under Architecture B.

The practical implication is direct. Dropout prevention, whether through shorter trial durations, participant engagement strategies, or adaptive scheduling, is far more consequential for the biomarker-interaction test than the choice of analysis-model carryover specification. Trialists who invest in refining the carryover specification while tolerating high dropout rates have, in our view, the priorities somewhat reversed.

5.6 Architecture dependence

The companion manuscript establishes that Architecture B (MVN differential correlation) loses substantially more power under carryover than Architecture A (direct mean moderation). The present results confirm that this qualitative gap is robust to decay-form variation: Architecture B is more sensitive to carryover under all three decay forms and at all five Weibull shape values examined. The A2 advantage is larger under Architecture B than under Architecture A, which is consistent with the mechanism. Under Architecture B, carryover erodes the biomarker-response correlation directly, and the continuous exposure-weighted predictor recovers more of that attenuated signal than either the binary on-drug indicator or the lagged-nuisance term can. The $A2 > A1 \approx A3$ ranking holds under both architectures, though the magnitude of the benefit warrants closer attention in trials where the data-generating mechanism is presumed to follow Architecture A.

5.7 The latent-subtype question

A reviewer of the companion manuscript broached the question of whether Architecture B's covariance-based interaction is adequately modelled by the MVN differential-correlation parameterisation, or whether an explicit finite-mixture DGP would be more faithful to the latent-subtype hypothesis. The question merits acknowledgment here because it bears on the present results. A true mixture DGP would produce biomarker-response associations that are qualitatively different from those generated by a continuous correlation: discrete responder and non-responder subgroups rather than a graded covariance structure. Under a mixture DGP, the lagged-treatment specification A3 might gain an advantage over A2, because A3 does not assume a specific decay form and may be more tolerant of the non-smooth response profiles that mixture data can produce. The present simulation does not evaluate a mixture DGP and cannot address this question directly. We have not been able to determine, as yet, whether the A2 dominance we observe extends to genuinely discrete-subtype data-generating processes; this is a direction that merits investigation in subsequent work.

5.8 Comparison with prior simulation studies

Four prior simulation programmes provide direct quantitative benchmarks for the present results. We discuss each in turn, focusing on cells where the prior parameter ranges overlap with ours and noting where the design targets differ.

Hendrickson² introduced the biomarker-by-treatment interaction question for aggregated N-of-1 trials and reported power for the matched-decay (A2) analysis under exponential carryover at $t_{1/2} = 1$ week, $N = 70$, $c_{bm} = 0.45$, Hybrid design. The reported power was 0.82 at 1{,}000 replicates under a single architecture. The present 500-replicate factorial reproduces this result at the same cell to 0.860 ± 0.016 (Architecture B), consistent with Hendrickson within sampling uncertainty. The principal methodological extension here is the addition of A1 and A3 as comparators; Hendrickson did not contrast analysis-side specifications, so the A2-versus-A1 advantage of approximately 10 percentage points is a new contribution rather than a replication.

Jones and Kenward³ developed the crossover-trial framework underlying the A3 specification with the expectation, noted above, that the non-parametric lagged-on indicator should outperform a parametric carryover model under decay-form mis-specification. The present S2 sensitivity block directly tests this expectation and finds it is not supported in the regimes we have examined: although A2's power varies by roughly 0.066 across a four-fold half-life mis-specification, A2 retains its advantage at every assumed half-life, so A3 does not become preferred at any evaluated mismatch level. The Jones-Kenward expectation is preserved in spirit, in that A3 closes the gap as half-life information degrades, but the practical magnitudes are too small for A3 to overtake A2 in the explored window. The disagreement is informative: it appears that the exposure-weighted predictor's curvature is gentle enough that mis-specification is less costly than the crossover-trial literature anticipates, at least at the prazosin-calibrated parameters.

Senn⁴ established the variance-decomposition foundation for within-subject designs, predicting that within-subject contrasts are disproportionately advantageous when between-patient variance $\sigma_{\text{between}}^2$ is large relative to within-patient variance. The present simulation does not vary $\sigma_{\text{between}}^2$ as a primary axis, but the S1 autocorrelation sensitivity block bears on the related question of how within-subject information accumulates with serial correlation. Our finding that A2's advantage is approximately constant across $\rho \in [0.5, 0.9]$ is qualitatively consistent with Senn's prediction that within-subject methods extract roughly proportional information across correlation regimes; the absence of a strong ρ interaction with the analysis specification is the new result here.

Sturdevant and Lumley⁵ developed the two-tier reporting convention we have adopted: a principal factorial covering the main contrast, supplemented by marginal sensitivity blocks anchored at a reference cell. Their convention was designed for simulation studies whose main claim is a ranking or contrast rather than a precise power estimate, and our adoption of it is documented in the `simulation-grid-plan.md` design document. The two-tier structure proved

useful in practice: the Tier 1 factorial establishes the $A2 > A1 \approx A3$ ranking, while the Tier 2 blocks probe its robustness to single-axis perturbations. The extension to the biomarker-interaction estimand is, to the best of our knowledge, new.

The companion manuscripts at `analysis/report/01-dgp-mean-moderation-vs-mvn/` and `analysis/report/05-nof1-design-sensitivity/` provide complementary benchmarks: the first establishes the architecture-dependence of carryover-induced power loss confirmed here in the architecture-dependence subsection, and the second reports the effect of carryover on the Hybrid N-of-1 design's saturation profile confirmed in §4.4.

5.9 Limitations

Several limitations should be noted when interpreting these results.

First, the simulation draws its response-trajectory parameters from a single configuration calibrated to the Hendrickson (2020) prazosin-PTSD framework (Gompertz maxima, rates, and dispersions). The absolute power values are specific to this parameterisation. The ranking of analysis specifications is expected to be more portable, but we have not been able to verify this across alternative response profiles. Trialists working in therapeutic areas with substantially different trajectory shapes should regard the power estimates as illustrative rather than definitive.

Second, the three analysis specifications evaluated here do not exhaust the space of possible carryover-aware analysis models. Weighted analysis, within-subject contrasts, two-stage random slopes, and exclusion of contaminated observations (discussed in the companion manuscript's Section 6) are not evaluated. We have endeavored to select the three specifications that represent the most natural practical alternatives for the N-of-1 context, but a more comprehensive comparison would cross the present grid with these additional strategies, and others may merit consideration.

Third, the sensitivity blocks are one-factor-at-a-time perturbations around a single reference configuration. Unfortunately, joint perturbations, for example high ρ combined with high dropout, may produce interactions that the marginal blocks do not capture. The exploratory block S5 ($\rho \times$ carryover) provides one such joint check and does not reveal a qualitative interaction, which is reassuring, but the space of possible joint combinations is large and our exploration of it is necessarily incomplete.

Fourth, the Tier 1 grid omits the open-label (OL) design because OL lacks the off-drug contrast required by the A1 and A3 specifications. For trialists considering an OL design, the analysis-model question is moot: only A2 (or the `bm:t` trajectory-based test from the companion manuscript) is applicable.

Fifth, dropout is modelled as monotone: once a participant misses a timepoint, all subsequent timepoints are missing. In practice, intermittent missingness is

common in N-of-1 trials, particularly when participants have good and bad symptom days that influence measurement compliance. Intermittent missingness would presumably be less destructive than monotone dropout because it preserves some late-trial observations that contribute to the treatment contrast. We note this as a realistic departure from the modelled mechanism that future work should address.

The headline Tier 1 ranking is further supported by the high-precision rerun reported at the end of the Results section, which resolves the A2-versus-A1 gap to MC SE 0.020 with paired McNemar $p < 10^{-3}$ at the headline cells.

6 Conclusions

1. **A2 (exposure-weighted) is the recommended default.** Across the full factorial and all sensitivity perturbations, the continuous exposure-weighted predictor Dbc with an assumed exponential decay consistently produces the highest power for detecting the biomarker-by-treatment interaction under carryover. The advantage over the binary-treatment (A1) and lagged-nuisance (A3) alternatives is approximately 10 percentage points at the reference configuration and is robust to DGP decay-form variation, within-factor autocorrelation, and analyst-truth half-life mismatch.
2. **Half-life mis-specification does not displace A2.** A2 power varies by roughly 0.066 across a four-fold mis-specification of the analyst's assumed half-life (Block S2), but A2 retains its advantage over A1 and A3 at every assumed half-life examined. Trialists who have only an approximate pharmacokinetic estimate can adopt A2 with confidence that a mis-specified half-life does not cost A2 its ranking.
3. **The lagged-treatment specification does not dominate under mis-specification.** Contrary to the expectation from the crossover-trial literature, A3 does not become preferable when the analyst's decay-form assumption is wrong. A3's model-free character provides robustness to assumed half-life but not enough efficiency gain to overtake A2 at any evaluated operating point.
4. **Dropout is more consequential than the carryover specification.** Power degrades sharply with dropout rate under all specifications. At 20% dropout, all three specifications lose approximately half their power relative to the no-dropout condition. Trialists should prioritise dropout prevention over carryover-specification refinement.
5. **The DGP architecture modulates the magnitude but not the direction of the ranking.** Architecture B (MVN differential correlation) amplifies the benefit of A2 because carryover-induced correlation erosion makes the continuous predictor's information advantage more consequential. Under Architecture A (direct mean moderation), A2's advantage is smaller but still positive.

718 **6. Reporting recommendations.** Simulation studies evaluating biomarker-
719 moderated treatment effects under carryover should report: (a) the analysis-
720 model carryover specification used,
721 (b) the assumed decay form and half-life, (c) sensitivity of power estimates
722 to at least one alternative specification, and (d) the anticipated dropout
723 rate, which is the dominant source of power loss in the evaluated grid.

724 **7. Type-I error is well-controlled.** All three specifications maintain nomi-
725 nal or conservative type-I error rates across the full Tier 1 grid and the S4
726 effect-size curve, with no evidence of inflation under any DGP decay form,
727 architecture, or carryover level.

728 7 Conflict of interest

729 The authors declare no conflicts of interest.

730 8 Funding

731 This work received no specific funding.

732 9 Data availability statement

733 The data and code supporting the findings of this study are available in the
734 pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and
735 ADEMP-compliant cell-level summaries are versioned under `analysis/data/`;
736 simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript
737 are listed in the Reproducibility section below.

738 10 Reproducibility

739 This manuscript was rendered with R 4.4.x inside the project's `zzcollab`
740 Docker container (image hash recorded in `Dockerfile`; package set captured
741 in `renv.lock`). Principal packages used by the simulation pipeline are `nlme`
742 (continuous-time linear-mixed analysis with `corCAR1` residual correlation),
743 `parallel` (8-core `mclapply` for parameter-grid sweeps), `data.table` (the
744 `original-extended` simulation collection), `furrr` and `purrr` (the `tidyverse`
745 simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build.

746 **Random-number generator.** Each simulation driver sets `set.seed(20260507)`
747 at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`.
748 Per-replicate seeds are derived from the master seed as `master + rep_idx`.

749 **Data and code availability.** Production-grade simulation outputs are
750 at `analysis/scripts/carryover-sensitivity/output/01-factorial.rds`
751 (principal factorial) and `04-sensitivity.rds` (sensitivity blocks S1 through
752 S5). Cell-level summaries are at `analysis/data/02-grid-summary.rds`

and analysis/data/02-sensitivity-summary.rds. Driver scripts are
 01-run-factorial.R, 04-run-sensitivity-blocks.R, and 02-summarise-grid.R
 under analysis/scripts/carryover-sensitivity/. The simulation grid plan
 and design rationale are documented at analysis/scripts/carryover-sensitivity/simulation-grid-plan.
 which serves as the Morris ADEMP pre-registration. High-precision prototype re-
 run outputs are at analysis/data/quick-sim/02-carryover-replicates.rds
 (per-replicate) and 02-carryover-summary.txt (Morris ADEMP cell-level sum-
 mary). The manuscript source, paper-specific bibliography (references.bib),
 and SIM CSL file are at analysis/report/02-carryover-sensitivity/.

11 References

1. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*. 2019;38(11):2074-2102. doi:10.1002/sim.8086
2. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: Statistical methods and practical considerations. *Frontiers in Digital Health*. 2020;2:13. doi:10.3389/fdgth.2020.00013
3. Jones B, Kenward MG. *Design and Analysis of Cross-over Trials*. 3rd ed. CRC Press; 2014.
4. Senn S. Mastering variation: Variance components and personalised medicine. *Statistics in Medicine*. 2016;35(7):966-977. doi:10.1002/sim.6739
5. Sturdevant SG, Lumley T. Statistical methods for testing carryover effects: A mixed effects model approach. *Contemporary Clinical Trials Communications*. 2021;22:100711. doi:10.1016/j.conctc.2021.100711